VIT Vellore Institute of Technology **Invention Disclosure Format (IDF)-B** *Submitted by: Abhik Boxi (24BCE0741), Harshith Reddy (24BCE2479), Harsh Saxena (24BCE0226) - School of Computer Science and Engineering*	**Document No.**	
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1. Title of the invention

A Secure Multi-Tenant Retrieval-Augmented Generation System with Storage-Level Pre-Traversal Isolation, Deterministic Bounded Grounding, and Page-Coordinate Provenance Tracking

2. Field / Area of invention

Natural language processing; retrieval-augmented generation (RAG); multi-tenant cloud storage; vector database security; approximate nearest neighbor graph search (HNSW); large language model hallucination control; document information retrieval.

3. Prior Patents and Publications from literature

The following literature provides the background for the proposed document question answering system. The main gaps addressed by the invention are tenant-bound retrieval, deterministic refusal when evidence is insufficient, and traceable page-level citations.

3.1 Foundation systems and retrieval augmented generation

Basic RAG foundations - Lewis et al. [1] and Gao et al. [2]

These works describe the standard RAG pipeline, which combines retrieval of external text with language model generation. External evidence can improve answers on knowledge-intensive tasks. The invention adds an authenticated multi-tenant document workflow, a fixed relevance gate, and document and page coordinates to this basic pipeline. In a shared university server, these additions are intended to prevent one user from retrieving another user's private documents.

3.2 Filtered vector search and tenant access control

Filtered HNSW search - ACORN [3] and Chronis et al. [4]

The disclosure identifies loss of recall when metadata filtering is applied only after a graph search. Filter-aware graph search addresses this retrieval problem. The proposed system additionally derives the tenant identifier from a signed JWT and binds it to vector metadata and the search predicate, rather than accepting an unrestricted tenant identifier from the client. Related filtered search work is listed in references [5] and [19].

3.3 Grounding and answer quality

Self reflective and corrective retrieval - Self-RAG [6] and CRAG [8]

Self-RAG and CRAG use model-based assessment to review retrieved evidence or generated responses. The invention instead places a deterministic cosine similarity gate before generation: if the best candidate score is below 0.65, the system returns a fixed fallback response without calling the language model. Hallucination surveys [13, 21] provide further background on unreliable generation.

Evaluation and context management - RAGAS [7], FActScore [14] and Liu et al. [9]

RAGAS and FActScore assess answer quality and factual support. They provide evaluation methods rather than the complete multi-user server architecture proposed here. Research on long-context use [9] motivates a bounded context of the top five retrieved chunks. The invention also carries document identifiers and physical source page numbers into the answer citations.

3.4 Document security and layout preservation

Prompt injection and knowledge poisoning - Zou et al. [10] and Greshake et al. [11]

These works show that malicious text in retrieved material can influence model behavior. The proposed system treats uploaded text as source data, normalizes extracted content, and encloses retrieved passages within delimiter fences before prompt assembly. References [12] and [18] provide related robustness and application security context.

Document layout and inference engines - references [15 to 17, 20, 22]

The disclosure uses layout-analysis work to motivate preserving reading order and source coordinates during ingestion. It uses Sentence-BERT embeddings for retrieval, a local quantized LLaMA-3 backend through Ollama for sensitive documents, and a Gemini API backend for approved large-context tasks.

4. Summary and background of the invention (Address the gap / Novelty)

4.1 Background and the technical problem

A document question answering service must retrieve relevant evidence, protect private files belonging to different users, and show where an answer came from. The lab evaluation reported a 28.4 percent hallucination rate for a standalone language model without retrieval. During prototype development, three practical problems were identified:

- **Tenant data exposure.** When tenant identity is handled only through client parameters or application checks, incorrect filtering or parameter tampering can expose another tenant's files.

- **Generation without sufficient evidence.** A conventional RAG pipeline may still ask a model to answer when retrieval has found no relevant source passage.
- **Loss of document context.** Naive character-based extraction and chunking can disturb reading order, split sentences, and discard the page coordinates needed to verify an answer.

4.2 The proposed solution in short

- **Tenant identity bound to storage.** A signed JWT supplies the tenant identifier. Each stored vector carries metadata containing tenant_id, doc_uuid, chunk_idx, source_page and timestamp. The retrieval path applies the tenant predicate during graph search.
- **Deterministic relevance gate.** If the highest candidate cosine similarity is below tau = 0.65, the system returns "Information not found in source material." without invoking the language model.
- **Page coordinate provenance.** Chunks preserve document UUIDs, chunk indices and physical source page numbers. Generated answers use citations in the form [Document_UUID, Page_Number]. The reported citation precision is 98.2 percent.
- **Local and cloud inference.** Sensitive documents use an on-premises 4-bit quantized LLaMA-3 model through Ollama. Approved large-context tasks, including tasks exceeding 8,000 tokens, can use the Gemini API according to tenant privacy policy.

5. Objective(s) of Invention

The objectives of the invention are as follows:

- **Multi-format document support.** Build a decoupled multi-user question answering service that handles PDF and DOCX files while preserving tables, headings and reading order.
- **Tenant isolation.** Prevent tenant data from mixing by binding authenticated tenant identity to vector metadata and index predicates.
- **Context continuity.** Reduce loss of context at chunk boundaries using a sliding-window chunker with a 512-token capacity and a 64-token overlap.
- **Controlled refusal.** Suppress generation for insufficiently relevant retrieval results using the deterministic similarity threshold tau = 0.65.
- **Verifiable attribution.** Return document and page coordinates for generated factual answers.
- **Dual inference.** Keep sensitive document inference local while supporting a cloud backend for approved large-context tasks.
- **Prompt injection resistance.** Reduce the influence of malicious instructions embedded in uploaded documents through normalization and delimiter fencing.

6. Working principle of the invention (in brief)

A user authenticates through FastAPI and receives a signed JWT containing a tenant identifier. PDF documents are parsed using glyph coordinates to reconstruct reading order; DOCX files are parsed through their OpenXML document trees.

The extracted text is divided into 512-token windows with 64 tokens of overlap. Sentence-BERT converts each chunk into a dense vector. The vector is stored with the metadata record {tenant_id, doc_uuid, chunk_idx, source_page, timestamp}.

For an authenticated query, the system generates a query vector and searches using the tenant-bound predicate. It calculates the highest cosine similarity among the retrieved candidates. A score below 0.65 produces the fixed fallback response immediately, without a language model call.

When the score is at least 0.65, the top five chunks are normalized and enclosed in delimiter fences. The selected local or cloud inference engine generates an answer from this bounded context and includes the document UUID and page number in the citation.

7. Description of the invention in detail

The architecture combines authentication, document ingestion, overlapping chunking, dense vector retrieval, a deterministic grounding controller, and decoupled local or cloud generation.

Figure 1 described in words: system architecture overview

The client communicates with a FastAPI authentication and orchestration layer. An ingestion engine extracts text and coordinates from uploaded PDF or DOCX files. A chunking and embedding pipeline writes vectors and tenant metadata to ChromaDB. At query time, a tenant-bound retrieval controller obtains candidate passages, applies the numerical similarity gate, and assembles a bounded prompt. A dispatch layer selects the inference backend, and the response layer returns the answer with source page coordinates.

7.1 Layout preserving ingestion engine

PDF text is represented by glyph positions rather than explicit paragraphs. The ingestion pipeline examines glyph bounding boxes (x0, y0, x1, y1), identifies column gutters, and reconstructs reading order column by column. Form-feed boundaries track the source page. The disclosure cites DocLayNet [22] and Nougat [17] as layout-related background. For DOCX files, the OpenXML DOM tree is parsed to extract headings and available page-break information.

7.2 Sliding window chunking and vector representation

The normalized text stream T is divided into overlapping windows. Let S = 512 tokens be the window capacity and O = 64 tokens be the overlap; the step between window starts is S - O = 448 tokens. For zero-based chunk index i, the chunk interval is:

Chunk_i = [i * (S - O), i * (S - O) + S].

Sentence-BERT produces the dense representation v_chunk = f_embed(Chunk_i) in R^d, normalized so that ||v||_2 = 1. Each vector is associated with M_i = {tenant_id, doc_uuid, chunk_idx, source_page, timestamp}. The overlap preserves shared context between adjacent chunks and reduces the effect of boundary truncation.

7.3 Storage level tenant isolation in HNSW graphs

The authenticated tenant identifier is obtained from the signed token rather than a freely supplied query parameter. The disclosed retrieval mechanism uses the predicate:

Predicate(u) = (M_u['tenant_id'] == Token_tenant_id).

The proposed storage-level mechanism applies this condition during HNSW graph routing, before candidate distance evaluation, to restrict retrieval to the authenticated tenant. The disclosure reports retrieval latency of 14.2 ms and zero cross-tenant vector exposures in 50,000 synthetic test queries. These observations describe the reported test conditions; they do not establish an absolute security guarantee.

7.4 Deterministic similarity threshold fallback

The controller computes cosine similarity between the query vector q_vec and each candidate vector v_k:

Sim(q_vec, v_k) = (q_vec . v_k) / (||q_vec|| * ||v_k||); s_max = max_k Sim(q_vec, v_k).

- **Below threshold.** If s_max < 0.65, the controller terminates generation and returns "Information not found in source material." The language model is not invoked for that query.
- **At or above threshold.** If s_max >= 0.65, the controller permits bounded prompt construction from the retrieved evidence. The lab evaluation reports a hallucination rate below 1.5 percent for the complete system.

7.5 Context bounded prompt synthesizer and provenance format

The top five retrieved chunks are sanitized and enclosed in delimiter fences. The prompt directs the model to answer from the supplied evidence and return a concise answer with exact attribution. The response format is:

Answer Text | Citation: [Document_UUID, Page_Number]

The chunk metadata provides the document identifier and source page coordinate. Bounded context is intended to reduce distraction from long, weakly relevant passages, while delimiter fencing separates source material from system instructions.

7.6 Decoupled local and cloud inference engines

Sensitive documents are routed to an on-premises 4-bit quantized LLaMA-3 8B model through Ollama. Approved large-context or cross-document queries may be routed to Gemini API. The dispatch decision is governed by tenant privacy policy. Keeping the sensitive-document inference path local is intended to prevent document content from being transmitted to the cloud backend.

8. Experimental validation results

8.1 Results from the lab prototype

The reported evaluation used an Intel Core i7-13700H machine with 32 GB RAM and an NVIDIA RTX 4060 GPU. The dataset contained 500 documents totaling 1.4 million tokens and 1,200 test questions, including 300 intentionally unanswerable queries.

Approach / Baseline	Hallucination rate	Recall@5	Citation precision	Average latency
Keyword search (BM25)	N/A - extraction only	52.1%	100% - literal match	45 ms
Standalone LLM (zero-shot, no RAG)	28.4%	N/A - no retrieval	0.0% - no citations	1.80 s
Naive RAG (LangChain + unfiltered HNSW)	12.7%	81.4%	64.5% - chunk only	1.42 s
Secure multi-tenant RAG system	< 1.5%	93.8%	98.2% - exact page	1.15 s

The figures above are the reported lab measurements. Latency is the average reported for each approach; no percentile distribution is specified.

8.2 Ablation findings and isolation testing

- **Retrieval quality.** Dense retrieval achieved Recall@5 of 93.8 percent, compared with 52.1 percent for BM25. Removing the 64-token overlap reduced recall to 78.6 percent.
- **Grounding gate.** The disclosure reports a reduction from the standalone model's 28.4 percent hallucination rate to below 1.5 percent with the complete system and the 0.65 similarity gate.
- **Tenant isolation.** Across 50,000 synthetic test queries, the evaluation reported zero cross-tenant vector exposures.

8.3 Scope of the reported validation

The reported results concern a lab deployment. The system has not yet been deployed under active production traffic across multiple campus departments. Further validation should measure behavior under concurrent tenant workloads, adversarial inputs, layout variations and changes in document or query distribution.

9. What aspect(s) of the invention need(s) protection?

The following method, system and computer-readable medium claims describe the aspects for which protection is sought.

Claim 1 (Independent Method Claim)

A computer-implemented method for secure, context-grounded document question answering in a multi-tenant computing environment, comprising:

(a) authenticating an incoming client session request against a cryptographically signed tenant security token, and deriving an immutable tenant identifier directly from said token;

(b) parsing an unstructured digital document into a normalized text stream while extracting and preserving physical document coordinates including page numbers and layout structures;

(c) partitioning said normalized text stream into overlapping chunks using a sliding window having a defined token capacity and a defined overlap stride;

(d) embedding each of said chunks into a dense vector representation via a sentence-transformer bi-encoder;

(e) storing each dense vector representation into a multi-tenant vector database alongside a persistent metadata record comprising at least said tenant identifier, a unique document identifier, a sequential chunk index, and a physical source page number;

(f) upon receiving a natural language query from an authenticated client, computing a dense query vector and executing an approximate nearest neighbor search across a graph index, wherein said tenant identifier is enforced as a pre-traversal constraint at graph nodes prior to distance evaluation, thereby strictly confining search exploration to vectors belonging to said tenant identifier;

(g) evaluating a maximum similarity score among retrieved candidate vectors against a predetermined numerical relevance threshold;

(h) where said maximum similarity score fails to satisfy said numerical relevance threshold, deterministically terminating execution and returning a fixed refusal response without invoking a generative language model; and

(i) where said maximum similarity score satisfies said numerical relevance threshold, constructing a context-bounded prompt encapsulated in delimiter fences and dispatching said prompt to a generative language model to synthesize an answer grounded strictly within the retrieved candidate vectors coupled with an inline page-coordinate provenance citation.

Claim 2 (Dependent Claim)

The method of claim 1, wherein said sliding window has a token capacity S = 512 tokens and an overlap stride O = 64 tokens, establishing an overlapping block transition Chunki = [i*(S-O), i*(S-O)+S].

Claim 3 (Dependent Claim)

The method of claim 1, wherein said approximate nearest neighbor search is executed over a Hierarchical Navigable Small World (HNSW) graph index, and wherein the tenant predicate filter is evaluated during graph routing to prune unauthorized neighbor exploration before cosine similarity distance calculations.

Claim 4 (Dependent Claim)

The method of claim 1, wherein said predetermined numerical relevance threshold is calibrated to a cosine similarity value of tau = 0.65, below which generation is unconditionally suppressed.

Claim 5 (Dependent Claim)

The method of claim 1, wherein parsing said digital document comprises spatial coordinate analysis of character glyph bounding boxes to identify column gutters and reconstruct reading order across multi-column layouts.

Claim 6 (Dependent Claim)

The method of claim 1, further comprising dynamically routing said context-bounded prompt to either a local quantised inference engine hosted on-premises or a cloud-hosted multimodal inference API based on a tenant privacy tier policy.

Claim 7 (Independent System Claim)

A secure multi-tenant document question answering system comprising one or more physical processors and non-transitory computer-readable memory storing instructions to establish: (a) an authentication subsystem validating signed JWTs to derive tenant identifiers; (b) a document ingestion engine parsing multi-format files while capturing spatial layout hierarchies and page indices; (c) a sliding-window chunking module dividing text into overlapping token sequences; (d) an embedding engine generating normalized dense vectors; (e) a multi-tenant vector database enforcing tenant pre-traversal graph filtering; (f) an orchestration controller deterministically intercepting generation when similarity is below threshold and synthesizing context-bounded prompts with page citations above threshold; and (g) a decoupled inference engine supporting local and cloud LLMs.

Claim 8 (System Claim)

The system of claim 7, wherein said multi-tenant vector database implements ACORN-style multi-hop link traversal across HNSW graphs to prevent recall collapse under high predicate selectivity.

Claim 9 (System Claim)

The system of claim 7, wherein said orchestration controller enforces strict delimiter sanitization on extracted text to neutralize adversarial indirect prompt injections.

Claim 10 (Medium Claim)

A non-transitory computer-readable storage medium comprising instructions that, when executed by one or more processors, cause the processors to execute the steps of claim 1.

10. What is Technology Readiness Level of your invention?

The invention is assessed at TRL 7: system prototype demo in an operational environment. The FastAPI, PDF/DOCX ingestion, ChromaDB, similarity gate, and LLaMA-3/Gemini components were tested end to end across 500 documents and 1,200 queries. Multi-department production deployment has not yet been evaluated.

Research			Development			Deployment		
TRL 1	TRL 2	TRL 3	TRL 4	TRL 5	TRL 6	TRL 7	TRL 8	TRL 9
Basic principles observed	Technology concept formulated	Experimental proof of concept	Technology validated in a lab	Technology validated in a relevant environment (industrially relevant for key enabling technologies)	Technology demonstrated in a relevant environment (industrially relevant for key enabling technologies)	System prototype demonstration in an operational environment	System complete and qualified	Actual system proven in an operational environment (competitive manufacturing for key enabling technologies, or in space)
						✓		

11.1 Testbed setup and baselines

- **Testbed.** Use several isolated tenants with separate document collections and authenticated accounts. Include PDF and DOCX files with tables, headings, multi-column layouts and page boundaries. Exercise local and cloud inference under the applicable privacy policies.
- **Baselines.** Repeat comparison against BM25 extraction, a standalone language model, and naive RAG. Keep document sets, query sets and model settings fixed across comparisons and record all configurations.
- **Question sets.** Include answerable, intentionally unanswerable, cross-tenant, boundary-spanning and adversarial questions. Record the expected answer sources and page coordinates before evaluation.

11.2 Experiments and adversarial scenarios

- **Scenario 1: tenant access enforcement.** Attempt to alter tenant parameters, use invalid or expired tokens, retrieve another tenant's document identifier, and submit concurrent queries. Inspect returned passages and citations for unauthorized exposure.
- **Scenario 2: threshold calibration.** Test queries with similarity scores below, at and above 0.65. Verify that below-threshold cases return the fixed fallback without a model call. Measure false refusals and unsupported answers separately.
- **Scenario 3: layout and provenance.** Compare extracted reading order, tables, chunk boundaries and citations with the original rendered documents. Include source material that spans physical pages and test DOCX pagination under the rendering environment.
- **Scenario 4: prompt injection and inference routing.** Upload documents containing adversarial instructions and delimiter-like text. Measure attack success and answer support. Check that privacy-sensitive prompts use the local backend and do not reach the cloud API.

- **Scenario 5: concurrent workload.** Increase the number of simultaneous tenants and queries. Record throughput, resource consumption, average latency, tail latency and retrieval quality while tenants share the service.

11.3 Measurements and acceptance review

Repeat each experiment with documented settings and independent query samples. Report sample counts, failure examples and uncertainty alongside aggregate results. Track Recall@5, hallucination rate, citation precision, refusal accuracy, cross-tenant exposure, prompt injection success, backend routing correctness and latency. Compare the repeat measurements with the lab values in Section 8; set deployment acceptance criteria before the pilot and review observed failures before increasing the readiness level.

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